

Immersive Chemistry Lab - IP2a Testing Plan

Testing Plan for Interactive Prototype 2a

DECO2300/7230 Digital Prototyping and Extended Reality

3-Sentence Project Pitch

Immersive Chemistry Lab is an XR chemistry learning prototype that connects a visible laboratory reaction with its molecular representation. Users manipulate two reactants, trigger a reaction, use a two-handed expansion gesture to move into molecular scale, and then directly construct and manipulate an H₂O molecule. The prototype uses spatial interaction, feedback and scale change to help make the relationship between observable chemistry and molecular structure easier to explore.

Testing Objective

The main purpose of this IP2a test is to evaluate whether the changes made after IP1 testing have improved the usability and clarity of the interaction flow. IP1 showed that users generally understood the basic lab setup and molecular manipulation, but controller mapping, grab distance, the lab-to-molecular transition, and lack of guidance caused repeated confusion.

This test therefore focuses on whether the revised prototype reduces those problems. I am not testing whether VR improves chemistry learning. I am testing whether the updated interaction design is easier to understand, whether feedback is noticed, and whether the lab and molecular stages still feel like one connected experience.

Testing question	What I want to learn
Can users interact naturally without remembering which controller belongs to which object?	Whether allowing either controller to grab either reactant or Hydrogen removes the controller-mapping confusion found in IP1.
Does proximity highlighting make grab range easier to judge?	Whether users understand when an object is close enough to interact with and whether this reduces trial-and-error movement.
Is the molecular transition now easier to discover?	Whether the instruction and animated outward arrows communicate the two-handed expansion gesture without an explicit demonstration.
Do users understand the molecular task and the atoms involved?	Whether the H ₂ O instruction, Hydrogen/Oxygen labels and target structure make the task clearer.
Does the new interaction feedback communicate progress and completion?	Whether users understand the temporary BOND FORMED message and the final H ₂ O COMPLETE feedback.
Do the lab and molecular stages still feel connected?	Whether users understand that the molecular environment represents a closer view of the reaction they just performed.

Testing Methodologies

I will use a moderated, task-based usability test with think-aloud observation, followed by a short semi-structured interview. Participants will be given goals rather than the exact interaction steps. This is important because the main purpose is to find out whether the revised cues and feedback are understandable without the facilitator giving the solution away.

During each task, I will record what the participant tries first, where they hesitate, which controller they naturally use, whether they notice visual feedback, and whether prompting is required. If a participant becomes completely stuck, I will use progressively stronger prompts and record the level of assistance provided.

Prototype Description / Requirements

Laboratory-scale section

The participant begins at a virtual laboratory bench with two reactants and a reaction beaker. Either controller can now interact with either reactant. Reactants change to a highlight material when a controller enters the valid grab range. Moving both reactants into the reaction area triggers the reaction effect and activates the molecular focus.

Scale-transition section

After the reaction, a green molecular focus appears above the beaker. The prototype now displays the instruction 'EXPAND YOUR HANDS TO EXPLORE MOLECULAR SCALE' together with animated arrows moving outwards. The participant performs the two-handed expansion gesture to enter molecular scale.

Molecular-scale section

The molecular environment contains one Oxygen atom and two Hydrogen atoms. The atoms are labelled, and the user is shown the task 'BUILD H₂O - WATER / Connect 2 Hydrogen atoms to 1 Oxygen atom'. Either controller can grab either Hydrogen, and Hydrogen atoms highlight when a controller enters grab range. Hydrogen snaps into valid positions around Oxygen, a temporary 'BOND FORMED' message appears after the first successful connection, and 'H₂O COMPLETE / Water molecule formed' appears when both bonds are formed. The participant can then grab a bonded Hydrogen, stretch the bond and pull far enough to break it.

Current Testing Limitation

This catch-up testing is being conducted outside the scheduled Week 9 studio using Unity's XR Interaction Simulator rather than a Meta Quest headset. The results can therefore evaluate interaction clarity, sequencing, feedback and discoverability, but they should not be treated as evidence about final headset ergonomics, physical reach, comfort or real controller tracking. Headset-based testing should be repeated in a later studio.

Assumptions Being Tested

Assumption	Evidence that would support it	Evidence that would challenge it
Either controller can be used naturally.	Participant reaches with whichever controller is convenient and successfully grabs different objects without asking which hand to use.	Participant still searches for a specific controller, swaps hands repeatedly, or believes objects are hand-specific.
Highlighting communicates valid grab range.	Participant notices the colour change and uses it as a cue that the object can now be grabbed.	Participant does not notice the change, misunderstands it, or continues guessing distance.
The transition cue communicates expansion.	Participant responds to the text/arrows by moving both controllers apart without a full demonstration.	Participant treats the focus as a button/object to grab, or cannot identify the intended movement.
Molecular instructions clarify the task.	Participant identifies Hydrogen/Oxygen and understands that the goal is to build H ₂ O.	Participant is unsure what the atoms represent or what structure they are trying to make.
Bond/completion feedback communicates progress.	Participant recognises that one bond has formed and understands when the molecule is complete.	Participant ignores the feedback or is unsure whether the task has succeeded.
The two scales feel connected.	Participant explains that molecular mode represents a closer view of the reaction.	Participant experiences molecular mode as a separate or unrelated activity.

Data Collection Method

I will use one observation sheet per participant rather than relying on memory after several sessions. Completion time may be noted, but speed is not the main success measure. A participant who takes longer but correctly understands the interaction is more useful than a fast completion after being told what to do.

Measure	How it will be recorded
Task completion	Independent / prompt required / not completed
Controller choice	Which controller the participant naturally used and whether hand mapping caused confusion
Grab-range feedback	Whether the highlight was noticed and what the participant thought it meant
First action attempted	Short note on what the participant naturally tried first
Hesitation / errors	Pauses, unexpected actions, overshooting, repeated attempts or misunderstandings

Help required
Interpretation of feedback
Scale connection
Think-aloud comments

Number and type of facilitator prompts
What the participant thought BOND FORMED and H2O COMPLETE meant
Whether the participant connected molecular mode to the earlier lab reaction
Short useful quotes or paraphrased comments

Success / Validation Criteria

Because this is a catch-up test and the available participant number may be smaller than the original in-class target, I will interpret results cautiously. The revised interaction model will be considered provisionally successful if the majority of participants can complete the main sequence with no more than minor prompting and if the new cues reduce the specific problems identified in IP1.

Criterion	Target
Participant completes the reaction stage using either controller	Majority independently or with one minor prompt
Participant interprets the highlight as an interaction/grab-range cue	Majority recognise the purpose
Participant initiates molecular transition from the revised cue	Majority without full gesture demonstration
Participant understands the H2O task and atom labels	Majority can explain the goal correctly
Participant forms both bonds and recognises completion feedback	Majority with no more than one minor prompt
Participant discovers or understands stretch/break interaction	Majority without full demonstration
Participant explains lab -> molecular connection	Majority correctly connect the two stages
Critical usability failure	No single critical interaction fails for most participants

Testing Setup

1. Open the IP2a Unity scene and reset all reactants, molecular objects and feedback text to their starting state.
2. Check that the XR Interaction Simulator and both controllers respond correctly.
3. Confirm that either controller can grab either reactant and that only one object is grabbed at a time.
4. Confirm that proximity highlighting works for reactants and Hydrogen atoms.
5. Check that Reactant A and B trigger the reaction and that MolecularFocus activates.
6. Confirm that the transition instruction and animated arrows appear and that the expansion gesture enters molecular mode.
7. Confirm that atom labels, H2O instruction, snapping, BOND FORMED, H2O COMPLETE, bond stretching and bond breaking all work.
8. Prepare the observation sheet before starting each participant.
9. Tell the participant that the design is being tested, not their chemistry knowledge, and ask them to think aloud.
10. Explain only the simulator/controller controls needed to operate the prototype. Do not explain the chemistry-specific sequence unless a prompt becomes necessary.

Testing Process - approximately 5-7 minutes per participant

1. Introduction - approximately 30 seconds

Say: "This is an updated prototype of an immersive chemistry lab. Imagine you are using VR controllers to perform a chemistry experiment and then explore what happens at molecular scale. I am testing whether the updated interactions and feedback make sense, not whether you know chemistry. Please say what you are thinking while you use it."

Explain the minimum simulator controls required to move the controllers and activate the grab input, but do not explain which controller should be used, the expansion gesture, the highlight meaning, or the molecular task.

2. Task 1 - Start the reaction - approximately 60 seconds

Participant instruction: "You have two reactants and a reaction beaker in front of you. Try to start the experiment."

Observe which hand/controller they choose, whether they switch hands naturally, whether both reactants can be manipulated without controller-mapping confusion, whether the proximity highlight is noticed, whether the beaker is understood as the destination, and whether the reaction feedback is noticed.

After the reaction ask: "Did anything help you judge when an object was close enough to interact with?"

3. Task 2 - Enter molecular scale - approximately 45-60 seconds

Once MolecularFocus appears, say: "You now want to investigate what is happening inside this reaction. Show me what you would try next."

Do not explain the expansion gesture first. Observe whether the participant reads the instruction, notices the animated arrows, moves both controllers apart, tries to grab/touch the focus, or searches for another control.

If stuck, use prompts in order: Prompt 1: "Look at the cues around the focus." Prompt 2: "What movement do the arrows appear to suggest?" Only demonstrate the gesture if they still cannot proceed.

After transition ask: "What do you think this new environment represents?"

4. Task 3 - Build the molecule - approximately 90 seconds

Participant instruction: "Use what you see here to complete the molecule."

Observe whether they identify Hydrogen and Oxygen from the labels, understand the H₂O instruction, use either controller naturally, notice Hydrogen highlighting, move Hydrogen toward Oxygen, understand snapping, notice the first-bond feedback, and recognise the final H₂O COMPLETE message.

After completion ask: "How did you know you had successfully completed the molecule?"

5. Task 4 - Manipulate the completed molecule - approximately 45 seconds

Participant instruction: "Now explore what happens if you manipulate the molecule you just created."

Avoid explicitly saying 'break the bond'. Observe whether they grab a bonded Hydrogen, pull or rotate it, notice the bond stretching, continue pulling, and understand the bond disappearing.

Afterwards ask: "Why do you think the bond changed or disappeared?"

6. Short post-test questions - approximately 60-90 seconds

- Which interaction felt most natural?
- Which interaction was least obvious or most confusing?
- Did it feel like you could use either controller naturally, or did you still think particular objects belonged to a particular hand?
- What did the colour/highlight change on the objects mean to you?
- What did the arrows and instruction around the green focus tell you to do?
- Did the Hydrogen/Oxygen labels and H₂O instruction make the molecular task clear?
- What did BOND FORMED and H₂O COMPLETE communicate to you?
- Did the molecular environment feel connected to the reaction you performed earlier? Why or why not?
- What is the one change you would make before the next prototype?

IP2a Observation Sheet

Participant	Reaction	Either hand natural?	Highlight understood?	Transition	H2O task clear?	H2O complete	Break bond	Prompts	Lab -> molecular?	Main hesitation / key comment
P1	I/P/N	Yes/Partly/No	Yes/Partly/No	I/P/N	Yes/Partly/No	I/P/N	I/P/N		Yes/Partly/No	
P2	I/P/N	Yes/Partly/No	Yes/Partly/No	I/P/N	Yes/Partly/No	I/P/N	I/P/N		Yes/Partly/No	
P3	I/P/N	Yes/Partly/No	Yes/Partly/No	I/P/N	Yes/Partly/No	I/P/N	I/P/N		Yes/Partly/No	
P4	I/P/N	Yes/Partly/No	Yes/Partly/No	I/P/N	Yes/Partly/No	I/P/N	I/P/N		Yes/Partly/No	
P5	I/P/N	Yes/Partly/No	Yes/Partly/No	I/P/N	Yes/Partly/No	I/P/N	I/P/N		Yes/Partly/No	

Per-participant notes

P1: First action / useful quote / unexpected behaviour / assistance given:

P2: First action / useful quote / unexpected behaviour / assistance given:

P3: First action / useful quote / unexpected behaviour / assistance given:

P4: First action / useful quote / unexpected behaviour / assistance given:

P5: First action / useful quote / unexpected behaviour / assistance given:

Basis for This Testing Plan

This plan follows the same task-based structure used for IP1 so that the updated prototype can be compared against the earlier usability findings. The focus has shifted from broadly discovering problems in the interaction model to checking whether the specific IP1 issues - controller mapping, unclear grab range, transition discoverability and missing guidance/feedback - have been improved in IP2a. Results from this catch-up test will be used as evidence for Design Evaluation 2, with the simulator-only testing limitation reported transparently.